



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 27: Public Goods, Monopsony and Monopolistic Competition

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ICS2523 Microeconomics

Lecture 27

Last lecture, we finished studying **Externalities**

- Solutions to the problem of externalities

And we started with **Public Goods**

In today's lecture, we will finish with:

- Public Goods
- Monopsony
- Monopolistic Competition

Face-to-face Class Assistanship

Today, we have a face-to-face class assistantship at 14.50 (B22).

We will review the topics for test 5.

Public Goods

Market Failures

19. Market Failures: Public Goods

Recap

Public goods are **nonexclusive** and **nonrival**

A good is **exclusive** if it is easy to exclude individuals from benefiting from the good once it is produced. A good is **nonexclusive** if it is impossible (or costly) to exclude individuals from benefiting from the good

Problem?

A good is **nonrival** if consumption of additional units of the good involves zero social marginal cost of production

Problem?

Public Goods and Resource Allocation

Market Failures

20. Market Failures: Public Goods and Resource Allocation

Resource allocation problems involving public goods turn out to be quite different from resource allocation problems involving private goods.

Private markets are not a very good mechanism for allocating public goods (other social institutions –such as voting- are used)

20. Market Failures: Public Goods and Resource Allocation

Let's illustrate this mathematically:

There are 2 individuals 1 and 2 and two goods in the economy: *a public good* (B) and *x* (private good). Each individual has an income w_i , $i = 1, 2$

Both finance public good *x* by giving t_1 and t_2 , respectively. Provision of the public good is given by

$$B = t_1 + t_2$$

20. Market Failures: Public Goods and Resource Allocation

Each individual will have the following utility functions:

$$u_1(B, w_1 - t_1)$$

$$u_2(B, w_2 - t_2)$$

where

$w_i - t_i = x_i$ represents private good consumption by individual

20. Market Failures: Public Goods and Resource Allocation

Each individual will have the following utility functions:

$$u_1(\mathbf{B}, w_1 - t_1)$$

$$u_2(\mathbf{B}, w_2 - t_2)$$

where

$w_i - t_i = x_i$ represents private good consumption by individual

Activity 1

$$u_1(\mathbf{B}, w_1 - t_1) , u_2(\mathbf{B}, w_2 - t_2)$$

where $w_i - t_i = x_i$ represents private good consumption by i individual

How is nonexclusivity reflected in this model?

How is nonrivalry reflected in this model?

20. Market Failures: Public Goods and Resource Allocation

How can we find optimum t_1 and t_2 ?

$$\text{Max}_{t_1, t_2} a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

$w_i - t_i = x_i$ represents private good consumption by individual

a_i represents a social weight between both individuals

20. Market Failures: Public Goods and Resource Allocation

$$\mathcal{L}(t_1, t_2) = a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

Solving the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial t_1} = a_1 \frac{\partial U_1}{\partial B} \cdot \frac{\partial B}{\partial t_1} + a_1 \frac{\partial U_1}{\partial x_1} \cdot \frac{\partial x_1}{\partial t_1} + a_2 \frac{\partial U_2}{\partial B} \cdot \frac{\partial B}{\partial t_1} = 0$$

$$= a_1 \frac{\partial U_1}{\partial B} \cdot 1 + a_1 \frac{\partial U_1}{\partial x_1} \cdot (-1) + a_2 \frac{\partial U_2}{\partial B} \cdot 1 = 0$$

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_1 \frac{\partial U_1}{\partial x_1}$$

$$\frac{\partial B}{\partial t_1} = \frac{\partial(t_1 + t_2)}{\partial t_1} = 1, \quad \frac{\partial x_1}{\partial t_1} = \frac{\partial(w_1 - t_1)}{\partial t_1} = -1$$

20. Market Failures: Public Goods and Resource Allocation

$$\mathcal{L}(t_1, t_2) = a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

Similarly:

$$\frac{\partial \mathcal{L}}{\partial t_2} = a_1 \frac{\partial U_1}{\partial B} \cdot \frac{\partial B}{\partial t_2} + a_2 \frac{\partial U_2}{\partial B} \cdot \frac{\partial B}{\partial t_2} + a_2 \frac{\partial U_2}{\partial x_2} \cdot \frac{\partial x_2}{\partial t_2} = 0$$

$$= a_1 \frac{\partial U_1}{\partial B} \cdot 1 + a_2 \frac{\partial U_2}{\partial x_2} \cdot (-1) + a_2 \frac{\partial U_2}{\partial B} \cdot 1 = 0$$

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_2 \frac{\partial U_2}{\partial x_2}$$

$$\rightarrow a_1 \frac{\partial U_1}{\partial x_1} = a_2 \frac{\partial U_2}{\partial x_2}$$

20. Market Failures: Public Goods and Resource Allocation

Since

$$MRS_1 = \frac{\partial U_1 / \partial B}{\partial U_1 / \partial x_1} \text{ and } MRS_2 = \frac{\partial U_2 / \partial B}{\partial U_2 / \partial x_2}$$

Using one of the first order conditions:

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_1 \frac{\partial U_1}{\partial x_1} \quad / \cdot \frac{1}{a_1 \frac{\partial U_1}{\partial x_1}}$$
$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} = 1$$

20. Market Failures: Public Goods and Resource Allocation

$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} = 1$$

Since $a_1 \frac{\partial U_1}{\partial x_1} = a_2 \frac{\partial U_2}{\partial x_2}$

$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_2 \frac{\partial U_2}{\partial x_2}} = 1 \rightarrow MRS_1 + MRS_2 = 1$$

20. Market Failures: Public Goods and Resource Allocation

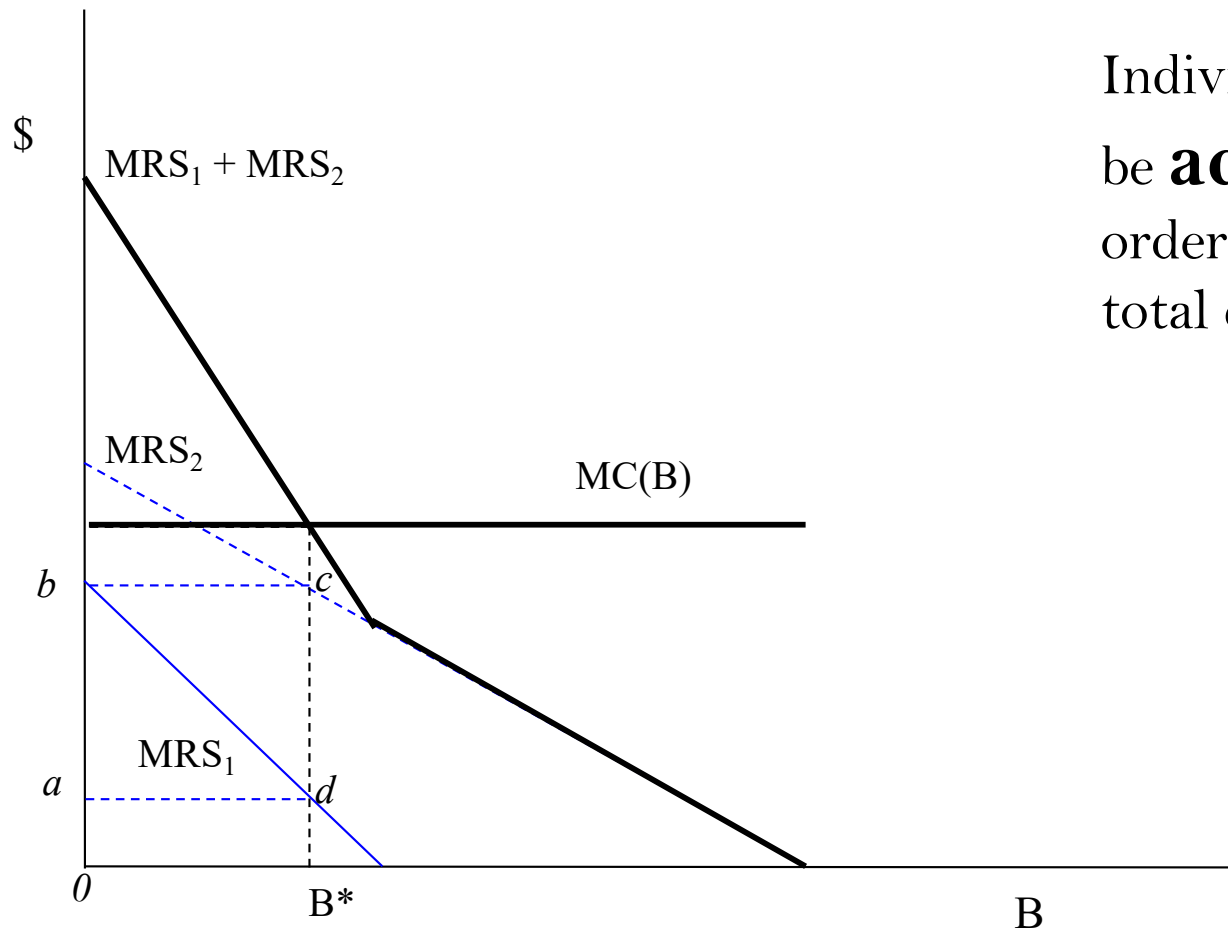
It is important to clarify that in this simple case, the **marginal cost** of providing public good B is 1.

Thus, the more general equation is that

$$MRS_1 + MRS_2 = MC(B)$$

The condition for efficiency in the case of continuous provision of the public good is that the sum of the marginal willingness-to-pay equals the marginal cost of provision

20. Market Failures: Public Goods and Resource Allocation

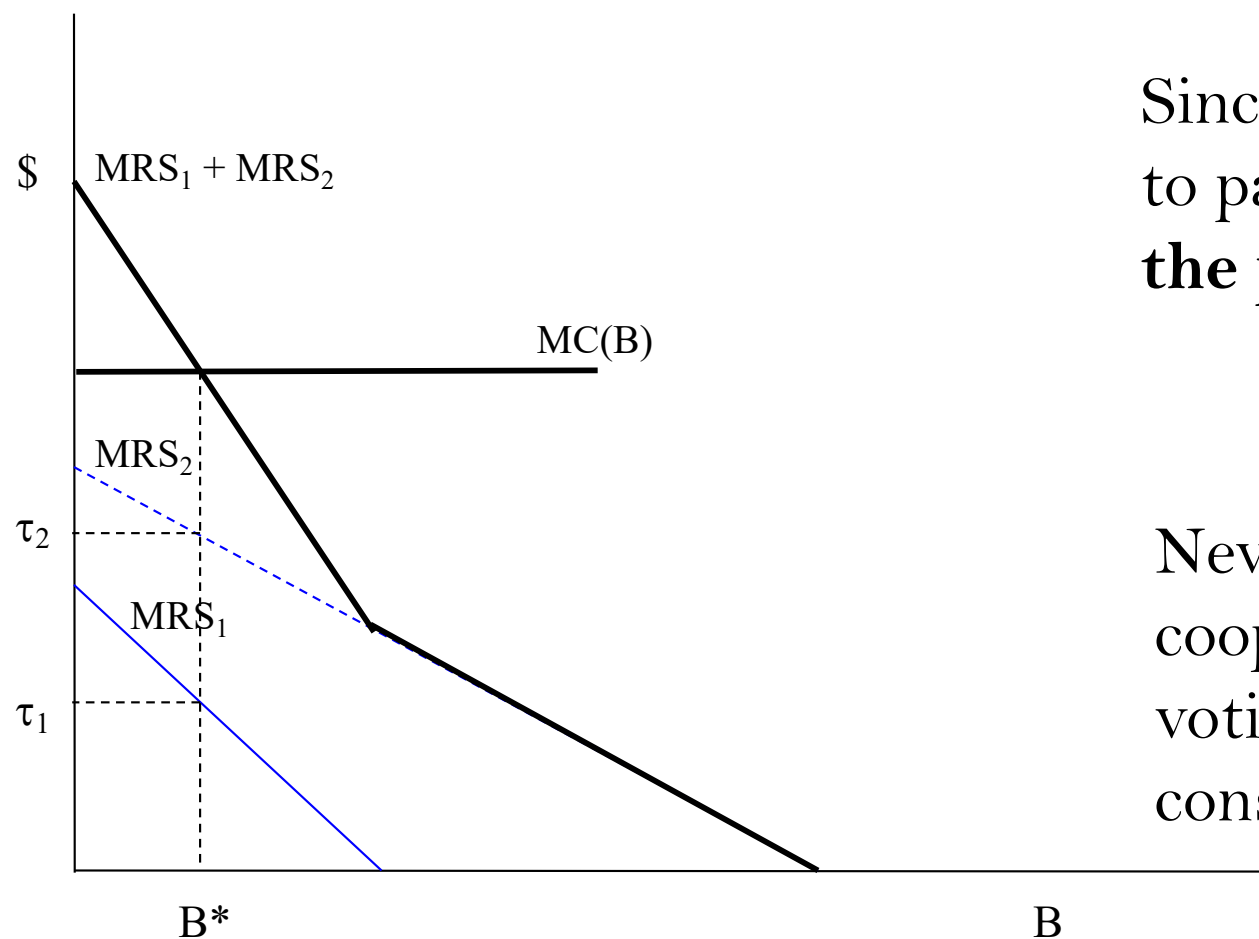


Individual demand curves must be **added vertically** in order to obtain public good's total demand.

Optimum provision quantity is B^* . Each individual should contribute to the financing of the public good the amount reflected in her demand curve for a B^* level of consumption

$$0adB^* = t_1; 0bcB^* = t_2$$

20. Market Failures: Public Goods and Resource Allocation



Since no one can be made to pay, **total provision of the public good is zero.**

Nevertheless, under a cooperation scheme or voting, the optimum level of consumption will be B^* .

20. Market Failures: Public Goods and Resource Allocation

Due to the nonrivalry and nonexclusivity, an individual knows that if the good is available, he can use it (without paying for it)

Thus, there is an incentive to reveal a lower willingness-to-pay than the real one → **free rider problem**

The market solution is an **insufficient provision of public goods**.

This is a market failure and we have to look up for alternative mechanisms of provision of public goods (voting systems, public provision → social evaluation of projects)

Efficient allocation
of public goods



Free riders
problem



Monopsony

Market Failures

21. Monopsony

A **monopoly** has a single seller, who chooses a price and quantity to maximise its profits.

A **monopsony** has a single buyer, who chooses a price-quantity combination from the industry supply curve that maximises its profits. It exercises its market power by buying at a price below what competitive buyers would pay.

An example: an American manufacturer of state-of-the-art weapon systems can legally sell only to the federal government, which is a monopsony.

21. Monopsony

Imagine that a firm is the sole employer in town (monopsony in the local labour market). The firm uses one factor, labour (L), to produce a final good.

The firm has a downward-sloping demand curve and faces an upward-sloping supply curve. The higher the daily wage (w), the more people are willing to work for the firm.

The firm's marginal expenditure (additional cost of hiring one more worker) depends on the shape of the supply curve. The supply curve shows the average expenditure (or wage) that the monopsony pays to hire a certain number of workers.

21. Monopsony

The monopsony's total expenditure is:

$$E = w(L) \cdot L$$

Its marginal expenditure is:

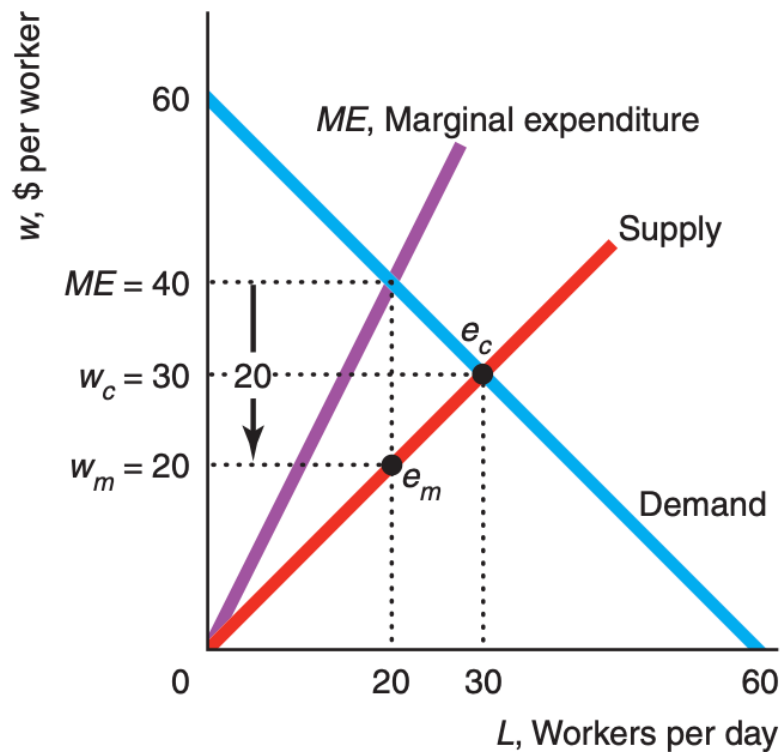
$$ME = w(L) + \frac{dw}{dL} \cdot L$$

Because the supply curve is upward sloping, $\frac{dw}{dL} > 0$, the marginal expenditure, ME, is greater than the average expenditure, $w(L)$.

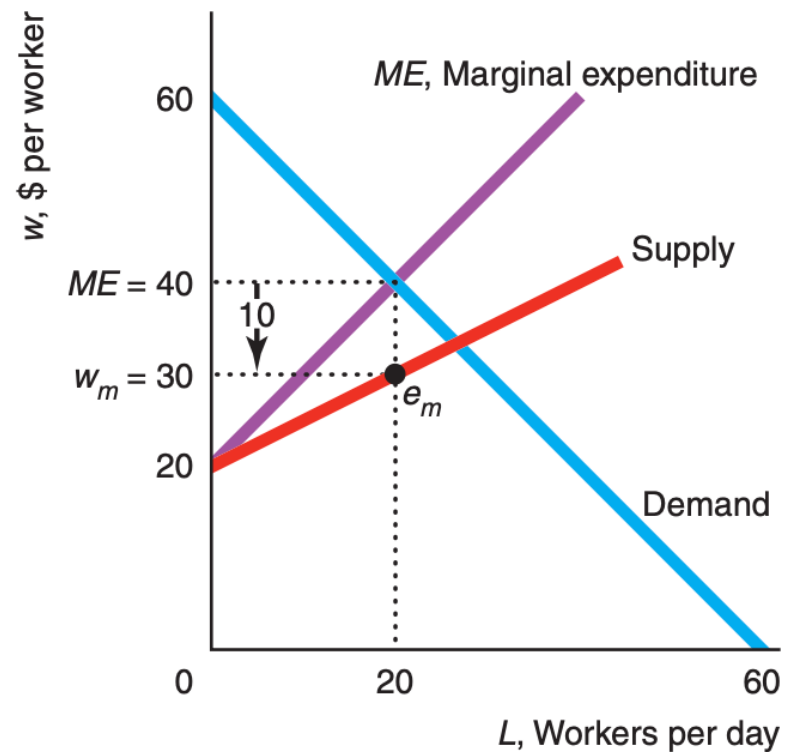
Any profit maximizing firm buys labour services up to the point at which the marginal value of the last unit of a factor equals the firm's marginal expenditure. If the last unit is worth more to the buyer than its marginal expenditure, the buyer purchases another unit. Similarly, if the last unit is less valuable than its marginal expenditure, the buyer purchases one less unit.

21. Monopsony

(a) Less Elastic



(b) More Elastic



21. Monopsony

We will assume that the firm is a price taker in the output market. It will choose how much labour to hire to maximise its profit:

$$\pi = P \cdot Q(L) - w(L) \cdot L$$

The first-order condition is:

$$\frac{d\pi}{dL} = 0$$
$$P \cdot \frac{dQ(L)}{dL} - w(L) - \frac{dw}{dL} \cdot L = 0$$

$$MR = \frac{dQ(L)}{dL} = w(L) + \frac{dw}{dL} \cdot L = ME$$

21. Monopsony

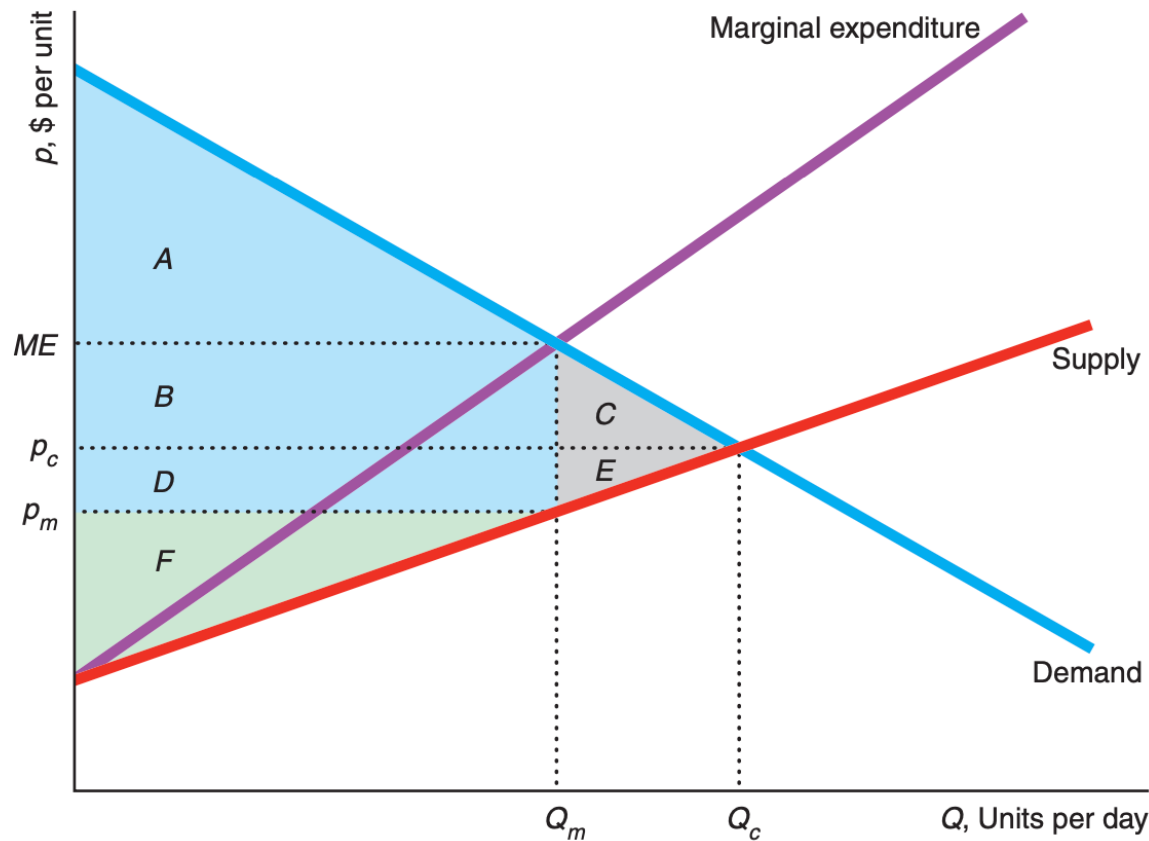
Monopsony power is the ability of a single buyer to pay less than the competitive price profitably. The size of the gap between the value the monopsony places on the last worker (the height of its demand curve) and the wage it pays (the height of the supply curve) depends on the elasticity of supply (η) at the monopsony optimum.

$$ME = w(L) + \frac{dw}{dL} \cdot L = w(L) \left(1 + \frac{dw}{dL} \frac{L}{w} \right) = w(L) \left(1 + \frac{1}{\eta} \right)$$

The markup of the marginal expenditure (which equals the value to the monopsony) over the wage is inversely proportional to the elasticity of supply at the optimum:

$$\frac{ME - w}{w} = \frac{1}{\eta}$$

21. Welfare effects of Monopsony



	Competition	Monopsony	Change
Consumer Surplus, CS	$A + B + C$	$A + B + D$	$D - C = \Delta CS$
Producer Surplus, PS	$D + E + F$	F	$-D - E = \Delta PS$
Welfare, $W = CS + PS$	$A + B + C + D + E + F$	$A + B + D + F$	$-C - E = \Delta W = DWL$

Monopolistic Competition

Market Failures

Imperfect competition

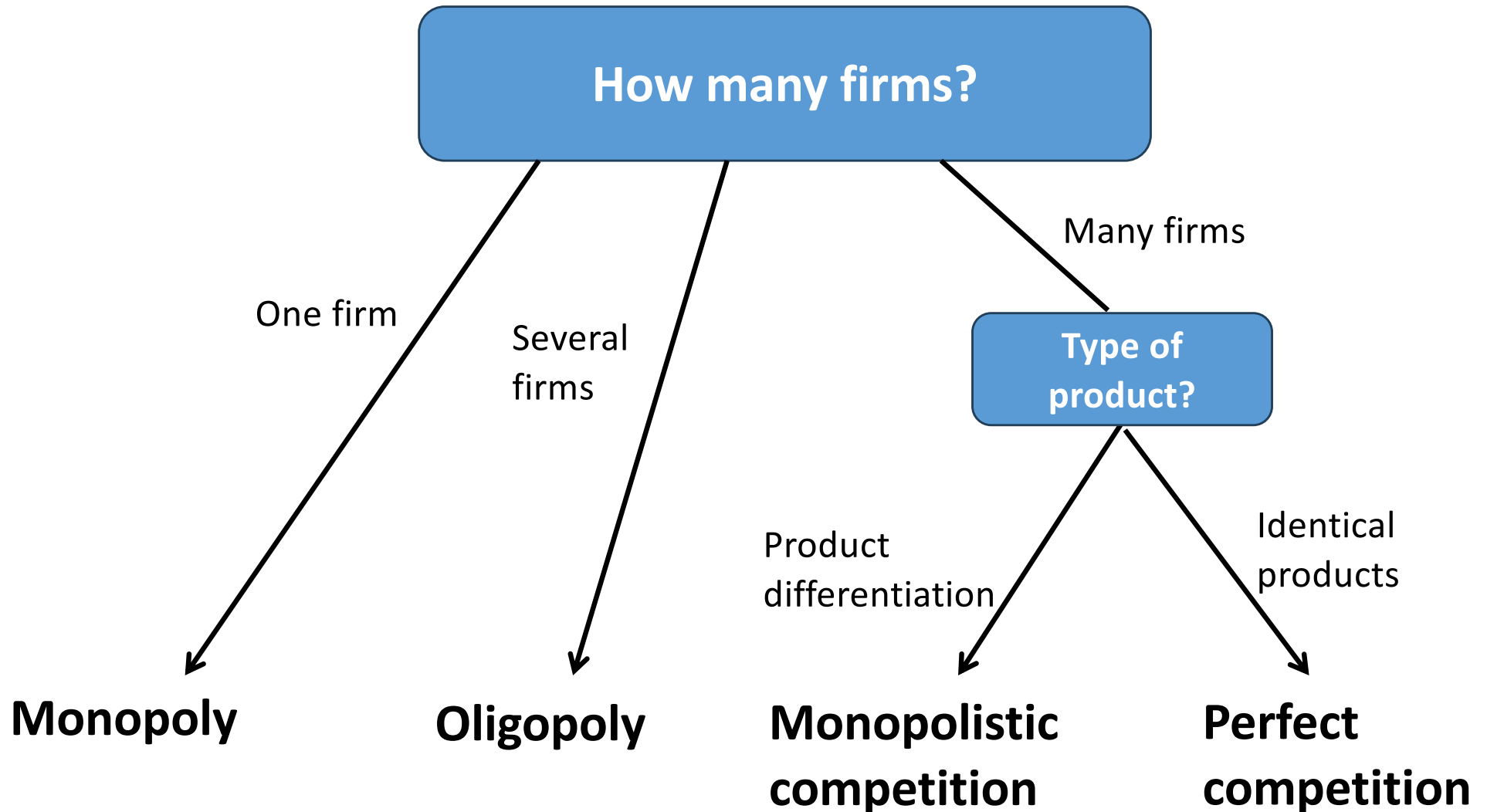
Other markets do not fit into a perfectly competitive market definition or a monopoly—they have elements of both types of markets and do not fit either model

→ Imperfect competition

One type of imperfect competition is **oligopolies**, a market structure in which few sellers offer similar or identical products —examples: cereals, detergent, OPEP, and cigarettes. Economists measure the market dominance exerted by a small number of companies with a statistic called the **concentration ratio**, which is the percentage of total output offered in the market by the four largest companies.

Another type of imperfect competition is **monopolistic competition**, a market structure where numerous companies sell similar, but not identical, products. These markets have three characteristics: many sellers, product differentiation, free entry and exit of the market —examples: computer games, restaurants, food trucks, cookies.

Imperfect Competition: Types of market structures



22. Monopolistic Competition

Monopolistically competitive firms **have market power** because they face downward-sloping demand curves, as do oligopolistic firms.

Video monopolistic competition:

<https://www.youtube.com/watch?v=T3F1Vt3IyNc>

However, in contrast to oligopolistic markets in which entry is very difficult, firms can freely enter a monopolistically competitive market, so **firms earn zero economic profit**, as do perfectly competitive firms

Competitive and monopolistically competitive firms make zero profits. Still, in contrast to competitive firms (which face horizontal residual demand curves and charge prices equal to marginal cost), **monopolistically competitive firms face downward-sloping residual demand curves**. Hence, they charge prices above marginal cost.

22. Monopolistic Competition

Monopolistically competitive firms face downward-sloping demand curves because the market is small or because the firms differentiate their products.

Equilibrium

Firms in monopolistic competition follow the profit maximisation rule of a monopoly:

- Optimal Quantity: They produce the quantity where marginal revenue (MR) equals marginal cost (MC).
- Price Determination: They use the demand curve to set the selling price for that quantity.

22. Monopolistic Competition

Equilibrium

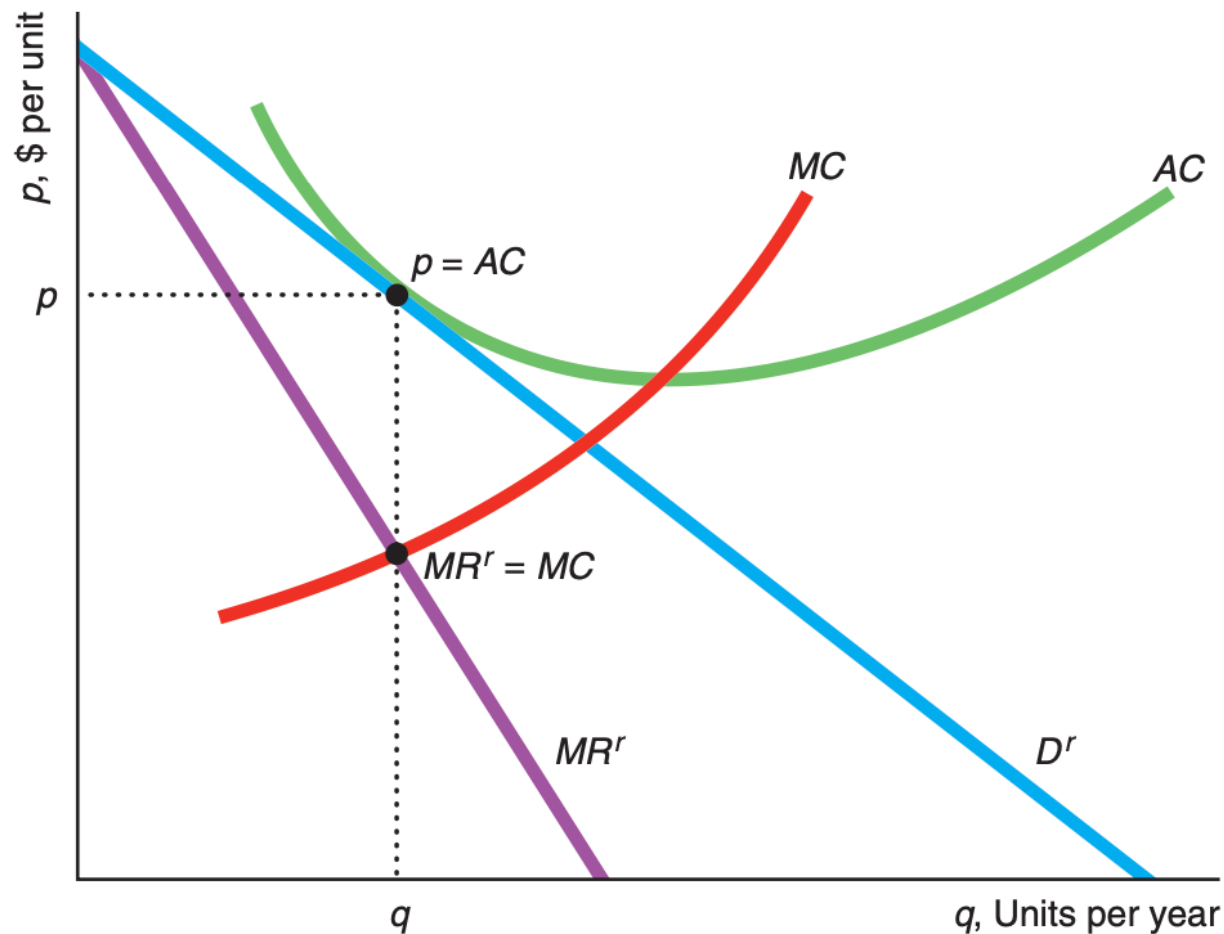
In the **short-run**, profits can be positive or negative.

In the **long-run**, the entry of new firms shifts the demand curve for existing firms to the left and makes it more elastic, as the supply of similar goods increases. In equilibrium, three conditions are met:

1. Each firm chooses its price and output level on its demand curve.
2. Each firm maximises its profits according to its demand curve.
3. The entry of new firms reduces profits to zero.

These facts imply the existence of a very special geometric relationship between the **demand curve and the average cost curve: both must be tangent.**

22. Monopolistic Competition



Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.